DAN MILL
TRAINING

OSI MODEL & TCP/IP MODEL CHEAT SHEET

OSI MODEL (7 LAYERS)

7 APPLICATION
6 PRESENTATION
5 SESSION
4 TRANSPORT
3 NETWORK
2 DATA LINK
1 PHYSICAL

TCP/IP MODEL (4 LAYERS)

4 APPLICATION
3 TRANSPORT
2 INTERNET
1 NETWORK ACCESS

OSI Memory Aid: "All People Seem To Need Data Processing"
Application → Presentation → Session → Transport → Network → Data Link → Physical

TCP vs UDP

TCP (Transmission Control Protocol):

- Connection-oriented
- Reliable delivery
- Error checking & correction
- Flow control
- Slower but guaranteed
- Used for: HTTP, FTP, Email

UDP (User Datagram Protocol):

- Connectionless
- Best-effort delivery
- No error correction
- No flow control
- Faster but no guarantee
- Used for: DNS, DHCP, Video

KEY DIFFERENCES

OSI Model:

- 7 layers (theoretical)
- More detailed separation
- Reference model
- Session & Presentation layers

TCP/IP Model:

- 4 layers (practical)
- Based on real protocols
- Implementation model
- Internet standard

COMMON PORTS

Port	Protocol	Service
20/21	TCP	FTP
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	UDP/TCP	DNS
67/68	UDP	DHCP
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
443	TCP	HTTPS

TROUBLESHOOTING GUIDE

Bottom-Up Approach:

- Physical: Check cables/power
- Data Link: Check switch/MAC
- Network: Check IP/routing
- Transport: Check ports/firewall
- Application: Check services

Tools by Layer:

- L1: Cable tester, multimeter
- L2: Switch LEDs, MAC table
- L3: Ping, traceroute, routing table
- L4: Netstat, port scanner
- L7: Browser, application logs

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